

Roll No. :

MID SEMESTER EXAMINATION SEPT. 2025

Name of the Program: **B.Tech (AIML and AIDS)**

Semester: I

Name of the Course : Introduction to Python Programming

Course Code: TCS102

Time : 90 Minutes

Maximum Marks : 50

Note:

- i. Answer all the questions by choosing any one of the sub questions
- ii. Each questions carries 10 marks

Q1	(10 Marks)	CO1
(a)	List and describe the five generations of computers in detail along with their main technologies used.	CO2
	OR	
(b)	Explain if-elif-else statement with the help of the following scenario: a. Input the three sides of a triangle from the user. b. Check and display whether the triangle is Equilateral, Isosceles, or Scalene. Draw a flowchart and develop a python program for the scenario provided.	
Q2	(10 Marks)	CO1
(a)	Describe lists and tuples in Python. Identify the key differences between them and illustrate at least 5 commonly used methods of lists and tuples each with examples.	CO2 CO3
	OR	
(b)	You are creating a project for managing marks. Develop a menu driven based python program to perform the following: a. Input the marks from the user and add it to the list. b. Display all the marks in the list. c. Find the highest and lowest marks and display. d. Find the total and average marks and display. e. Count how many students scored less than 40 marks. f. Increase the marks of every student by 5.	
Q3	(10 Marks)	CO1
(a)	Explain the different types of loops available in Python. Illustrate each type with suitable code snippets to demonstrate their working.	CO2
	OR	
(b)	Discuss try, except, else, finally, blocks in terms of exception handling. Develop a python program to take number of pizzas and number of friends as input from the user and find the number of pizzas each friend will get. Use try, except finally blocks and handle "ValueError" and "ZeroDivisionError" scenarios.	
Q4	(10 Marks)	CO1
(a)	Elaborate the memory hierarchy in a computer system with a suitable diagram and compare them in terms of cost, speed and time.	CO2
	OR	
(b)	Define a function in Python. Describe the different ways to pass arguments to a function and illustrate each with an example.	
Q5	(10 Marks)	CO1
(a)	You are a Python developer working on a text-processing project. You are developing a menu-driven string validator that performs various checks and transformations on user input.	CO2 CO3

	1. Check if a string contains all the vowels at-least once. Example: "Education is powerful" prints True Example: "Python coding" prints False 2. Format a string as a. Convert the string to lowercase b. Remove leading and trailing spaces(extra spaces at start and end of the string) c. Replace spaces between words with underscores (_). Example: " Hello World " and at last the output should be "hello_world" 3. Check if the first 3 characters and last 3 characters are the same using slicing. Example: "abcXYZabc" prints True (first 3 "abc" and last 3 "abc" are same). Example: "python" prints False	
	OR	
(b)	Implement a menu-driven Python program that allows the user to choose an operation. Create separate functions to: a. Count the number of digits in a given number b. Calculate the product of numbers up to n using recursion c. Find the nth Fibonacci number using recursion	